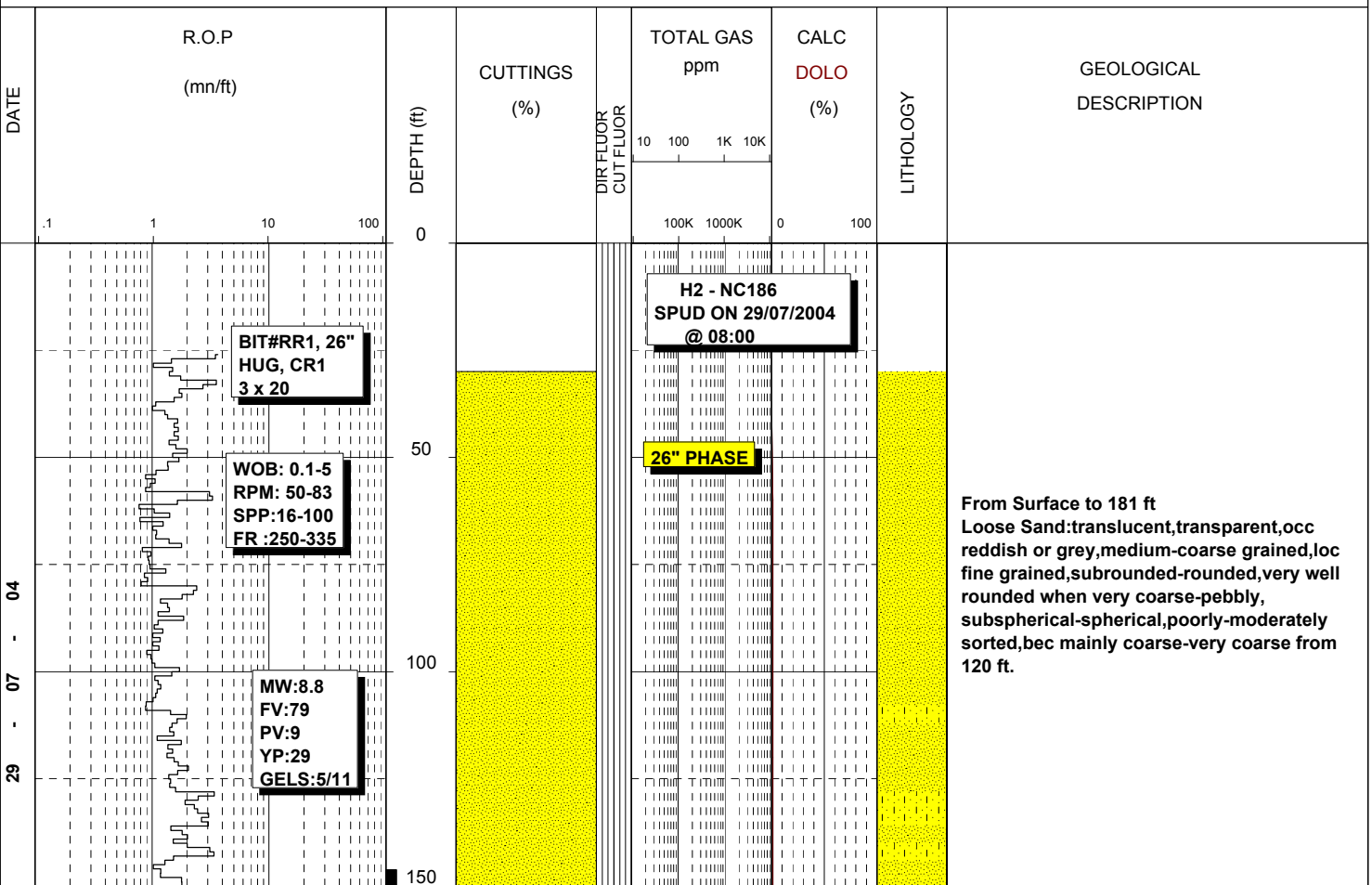
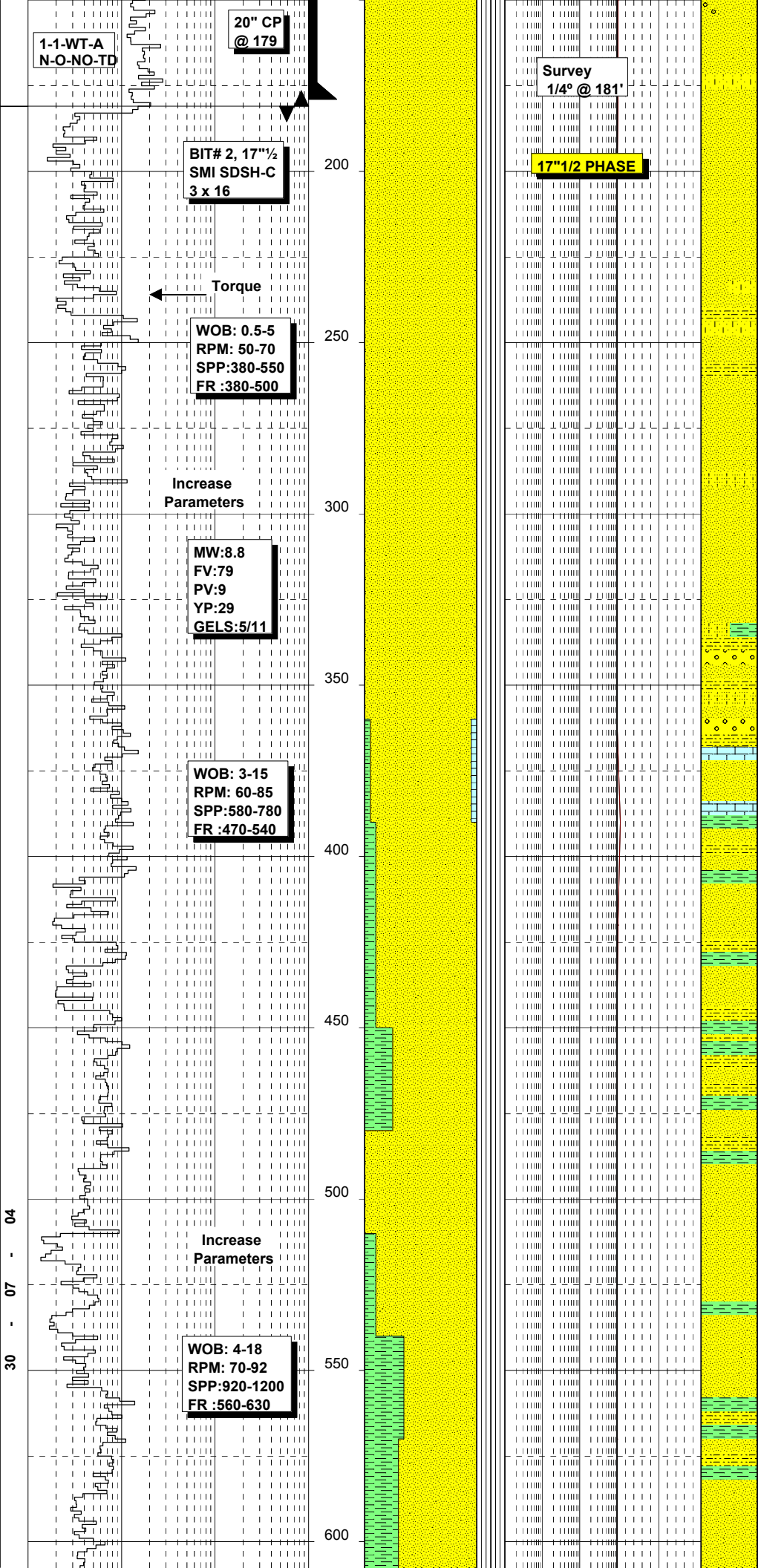
Ground level : 1734 ft
Kelly bushing: 1757 ft

Final TD :5450 5450

Generated by ALX Package

clay	Shale	MARL	Siltstone	Limestone
Slty,Sdy Cly/Sh	Sdy,Calc Cly/Sh	Slty MARL	Sand	Arg Limestone
Sdy Cly/Sh	Sdy,Dol Cly/Sh	Sdy MARL	Arg Sand	Sdy Limestone
Calc Cly/Sh	Anh Cly/Sh	Dol MARL	Arg Siltstone	Dol Limestone
Dol Cly/Sh	Pyr Cly/Sh	Sdy & dol MARL	Arg Sandstone	Slty Limestone
Cly Silicified	Varicol. Sh	Bit SHALE	Sandstone	Calc Dolomite
	Glauconitic	kaolinitic	Calc sandstone	Arg Dolomite
	Pyritic	Micaceous	Anhydrite	Dolomite
Test	Mud loss	Mud gain	Deviation surve	Core

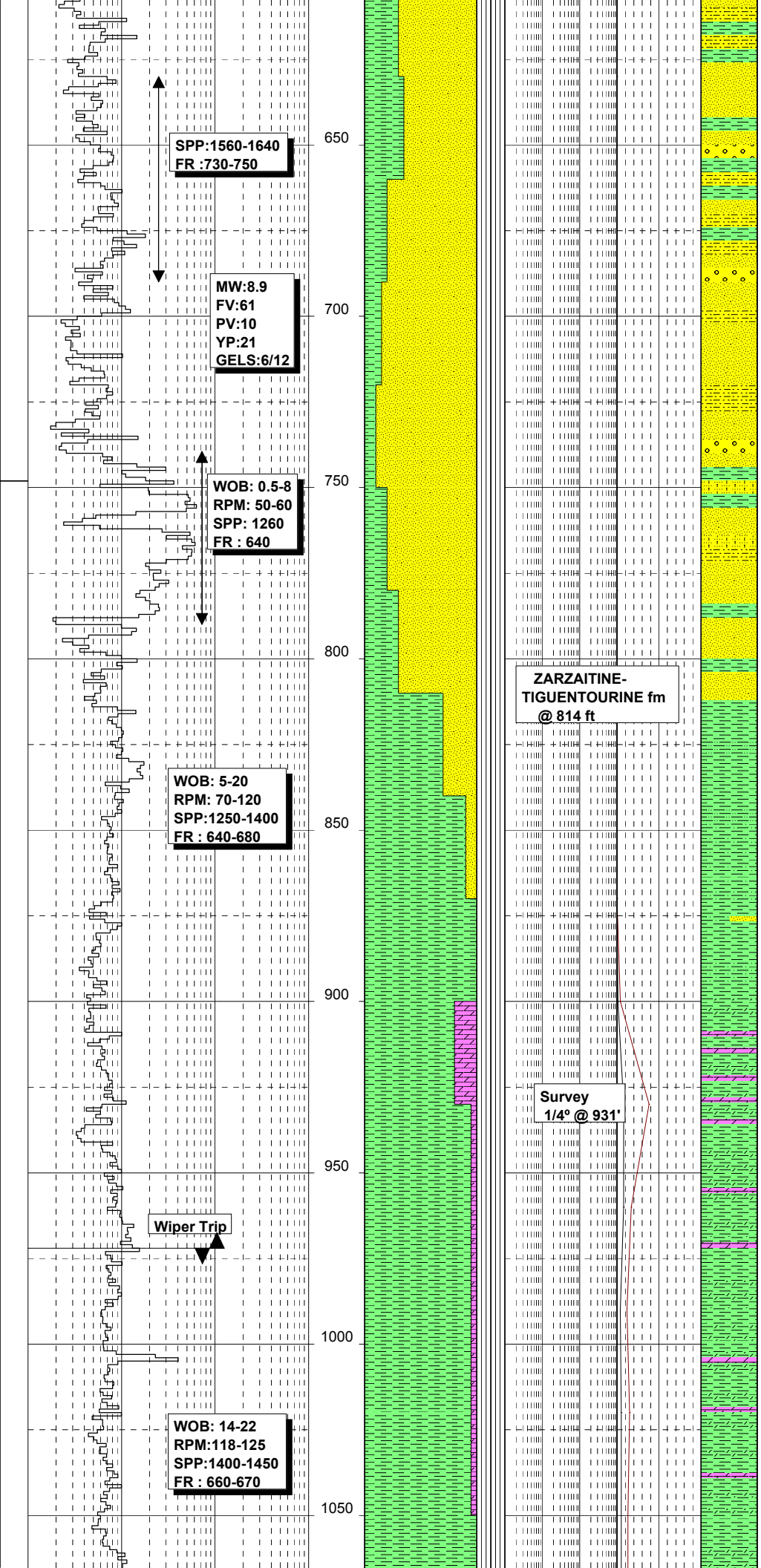




From 181 ft to 360 ft
Loose SAND:translucent,transparent,orange, fine upper-coarse grained,subangular-subrounded,well rounded when very coarse, moderately sorted,mainly spherical,occ subspherical to subelongated,grading to conglomeratic Sandstone with granules or pebbles from 330 ft,poorly consolidated with red or reddish white clayey matrix

From 360 ft to 480 ft
Loose SAND:a/a, mainly transparent,slightly orange,commonly conglomeratic.
CLAY:red brick,red,white or ochre i.p soft,washable,sandy,no calcareous.
Thin streaks at top of:
LIMESTONE:white,off white,micro-crystalline,soft-medium hard,commonly sandy grading to calcareous Sandstone.

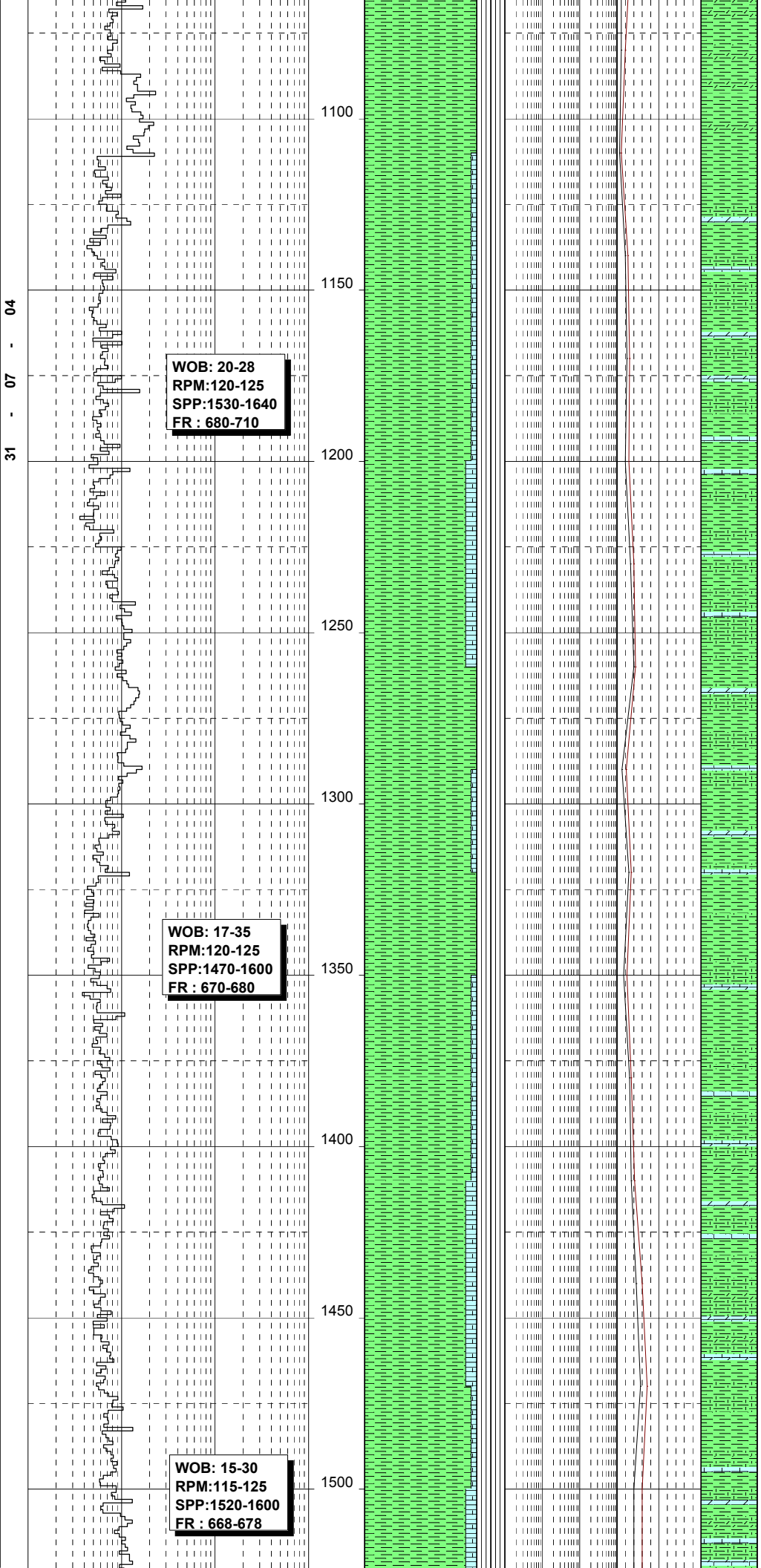
From 480 ft to 660 ft
Loose Sand:transparent,translucent, colorless,fine-medium,occasionally coarse grained,subangular-subrounded,moderately-well sorted.
CLAY:mainly red brick,locally off white or light grey,soft-firm,sandy in parts, rarely slightly calcareous,washable, grading to Claystone with depth.



From 660 ft to 814 ft
Loose SAND:as above,bec medium-coarse grained with abundant yellowish,brownish, rounded granules or pebbles (up to 1 cm), in parts slightly consolidated with light brown,ochre clayey matrix.
CLAY: as above.

From 814 ft to 900 ft
CLAY/CLAYSTONE:red brick,locally off white,soft to firm,blocky,finely sandy or silty,non calcareous,very slightly dolomitic i.p,grad locally to highly argillaceous Sandstone.
Minor Loose SAND:transparent,translucent, colorless,fine upper to medium grained, moderately-well sorted,rounded-subrounded.

From 900 ft to 1110 ft
CLAYSTONE:red brick,red brown,brown, commonly off white,very light green,silty-sandy i.p. at top,firm,blocky,i.p. fairly dolomitic w/ very fine dolomite crystals when brownish,and thin streaks of DOLOMITE:off white,very light cream,light grey,medium hard,micro-cryptocrystalline, argillaceous,locally earthy.



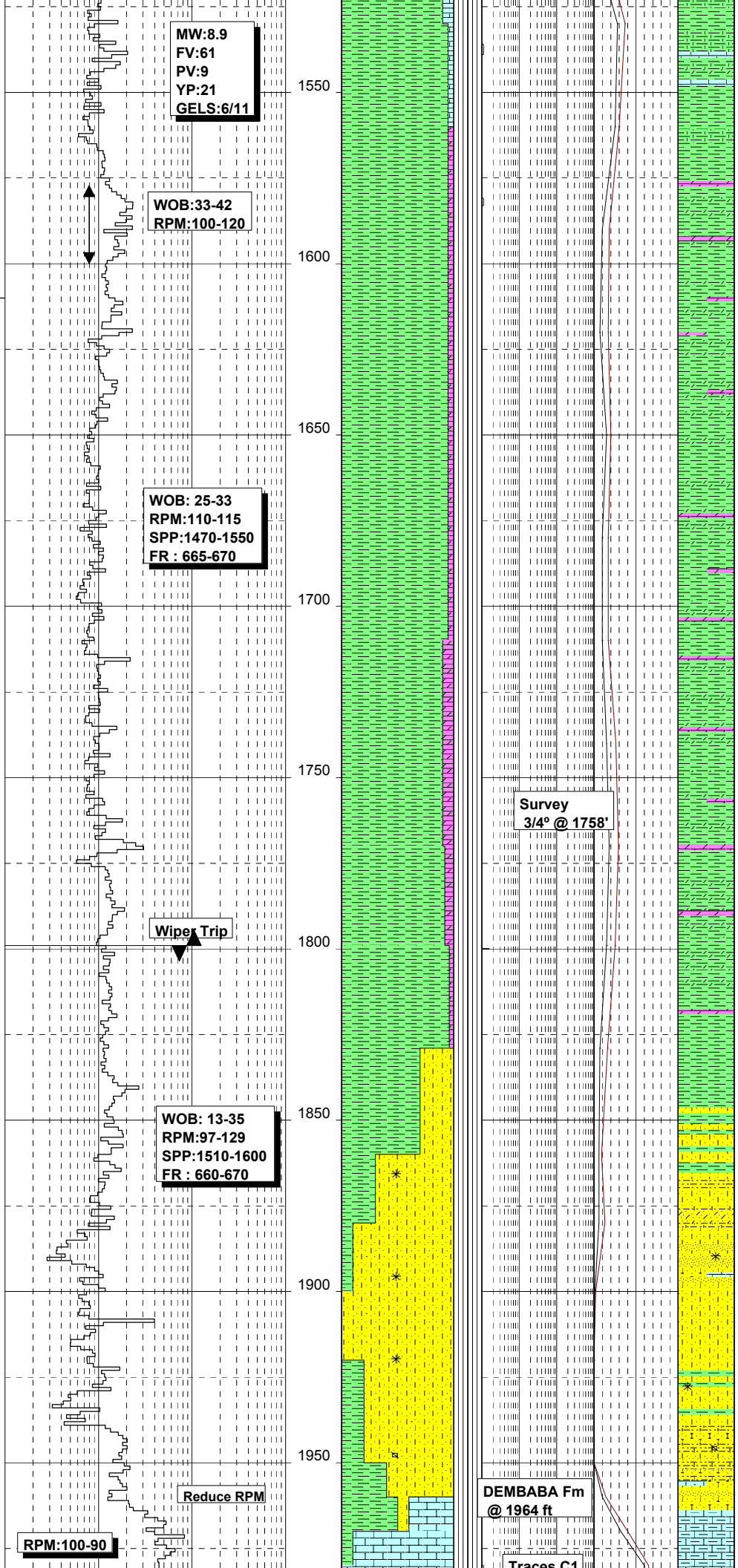
From 1110 ft to 1260 ft
CLAYSTONE:red brick,red brown,brown,occa-sionallylight green,very light grey,firm-well indurated,blocky,slightly-fairly calcareous with locally calcite crystals, slightly dolomitic i.p.
LIMESTONE:very light green,off white,cream medium hard,rarely soft and chalky,micro-cryptocrystalline,dolomitic grading to Dolomite.

From 1260 ft to 1350 ft
CLAYSTONE:brownish ochre,red brick,soft-firm,occ well indurated,blocky,slightly calcareo-dolomitic if reddish,fairly calcareo-dolomitic if brownish,common calcite crystals or veinlets.
LIMESTONE:cream,hard,cryptocrystalline and off white-white,medium hard,micro-very fine crystalline.

From 1350 ft to 1440 ft
CLAYSTONE:as above,more ochre,buff, purple i.p,firm,blocky-subblocky, highly calcareous i.p,very locally finely sandy.
LIMESTONE:as above.

From 1440 ft to 1500 ft
CLAYSTONE:mainly light brown,buff,second reddish purple,soft-well indurated,blocky, highly calcareous in parts.
LIMESTONE: as above.

From 1500 ft to 1560 ft
CLAYSTONE:red brick, sligthly purple, occasionally brownish,firm-well indurated, blocky,occasionally subblocky,highly calcareo-dolomitic.



Dolomitic LIMESTONE:white,light cream, very light green,medium hard,micro-crypto crystalline.

From 1560 ft to 1800 ft
CLAYSTONE:mainly brownish red,occ red brick,well-moderately indurated, blocky, slightly-fairly dolomitic, occasionally brittle w/ dolomite crystals
DOLOMITE:cream,off white,tan,medium hard, micro-fine crystalline,occasionally brownish,fine crystalline,highly argillaceous.

From 1800 ft to 1830 ft
CLAYSTONE:red brick,red brown,occ off white,ochre,well indurated,blocky-sub-blocky,sligthly-highly dolomitic
DOLOMITE:off white,very light green,cream medium hard,cryptocrystalline.

From 1830 ft to 1880 ft
CLAYSTONE: as above
SANDSTONE: brown,buff, reddish brown, friable-medium hard,fine-medium grained, subrounded-subangular,consolidated w/ abundant argillaceous cement,bec more clean,off white,occasionally transparent, friable,mediumgrained,then downwards brownish red,highly argillaceous.

From 1880 ft to 1964 ft
SANDSTONE: mainly loose,translucent-transparent,yellowish in parts,medium grained,subangular-subrounded in parts, locally slightly consolidated w/ silico-argillaceous cement,slightly glauconitic, becoming off white,tan,friable-medium hard,fine upper-medium grained,subounded, well sorted,consolidated w/calcareous cement, traces of pyrite.

From 1964 ft to 2033 ft
LIMESTONE:light buff,light cream,tanish, occasionally off white,medium hard,micro-

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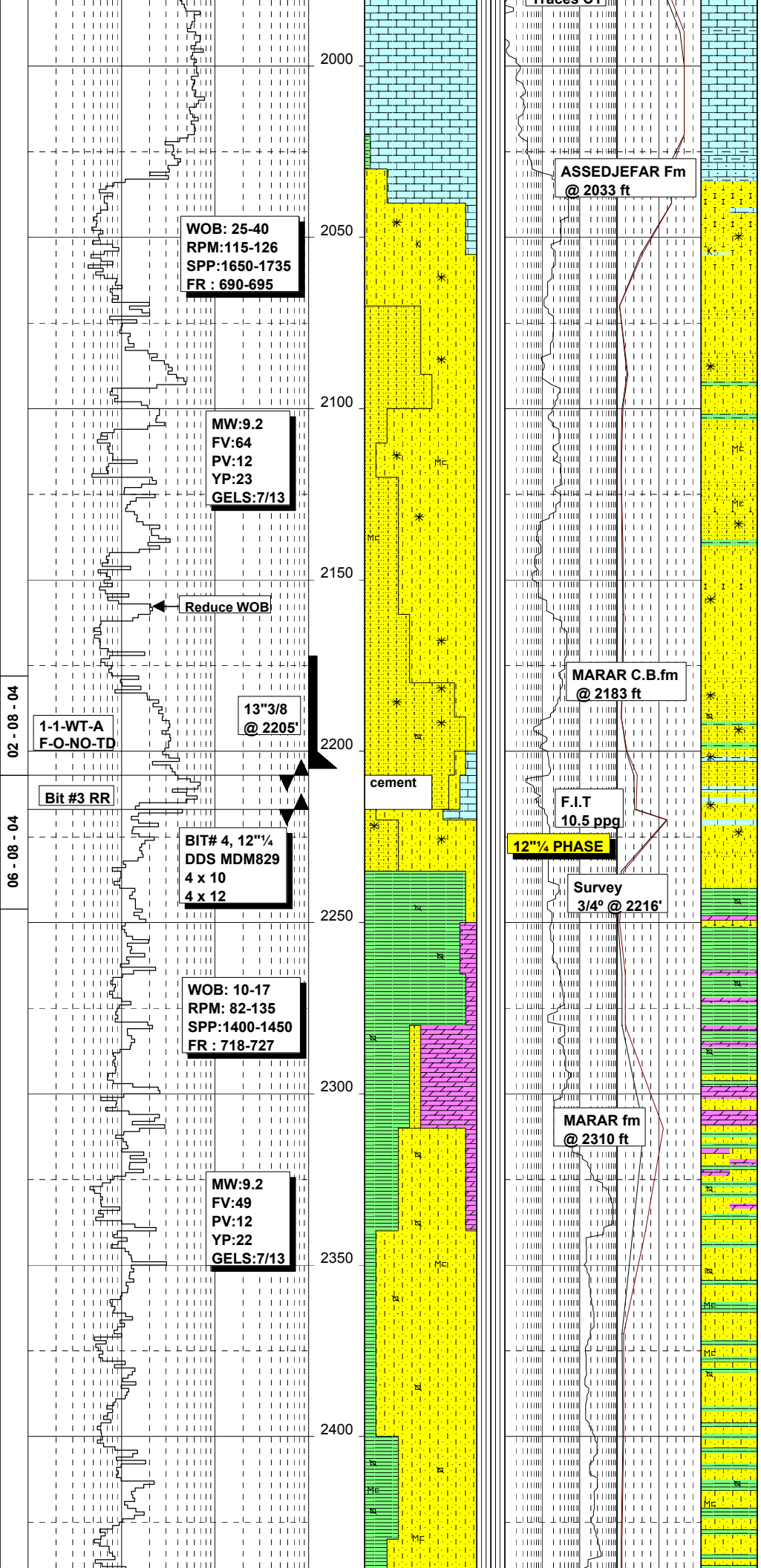


very fine crystalline,fairly-loc slightly argillaceous,sandy & glauconiticat top, becoming i.p off white,soft,chalky,occ bioclastic(fossil fragments).

From 2033 ft to 2080 ft
SANDSTONE:transparent,off white,very light cream,pinkish,friable,fine upper grained, well sorted,subangular,fairly-downwards poorly consolidated w/calcareous cement, locally kaolinitic,slightly glauconitic.

From 2080 ft to 2183 ft
SANDSTONE:tan,cream,medium hard-friable, commonly loose at top,fine-medium grained, subrounded-subangular,consolidated w/ calcareous cement,slightly glauconitic, rarely micaceous.
SILTSTONE:grey,dark grey,medium hard, highly argillaceous,w/numerous spherules (0.3mm size,micro fossils?),grading to silty Shale,occasional dark green free glauconite,finely pyritic i.p.
Traces of
LIMESTONE:off white,light cream,hard, recrystallised

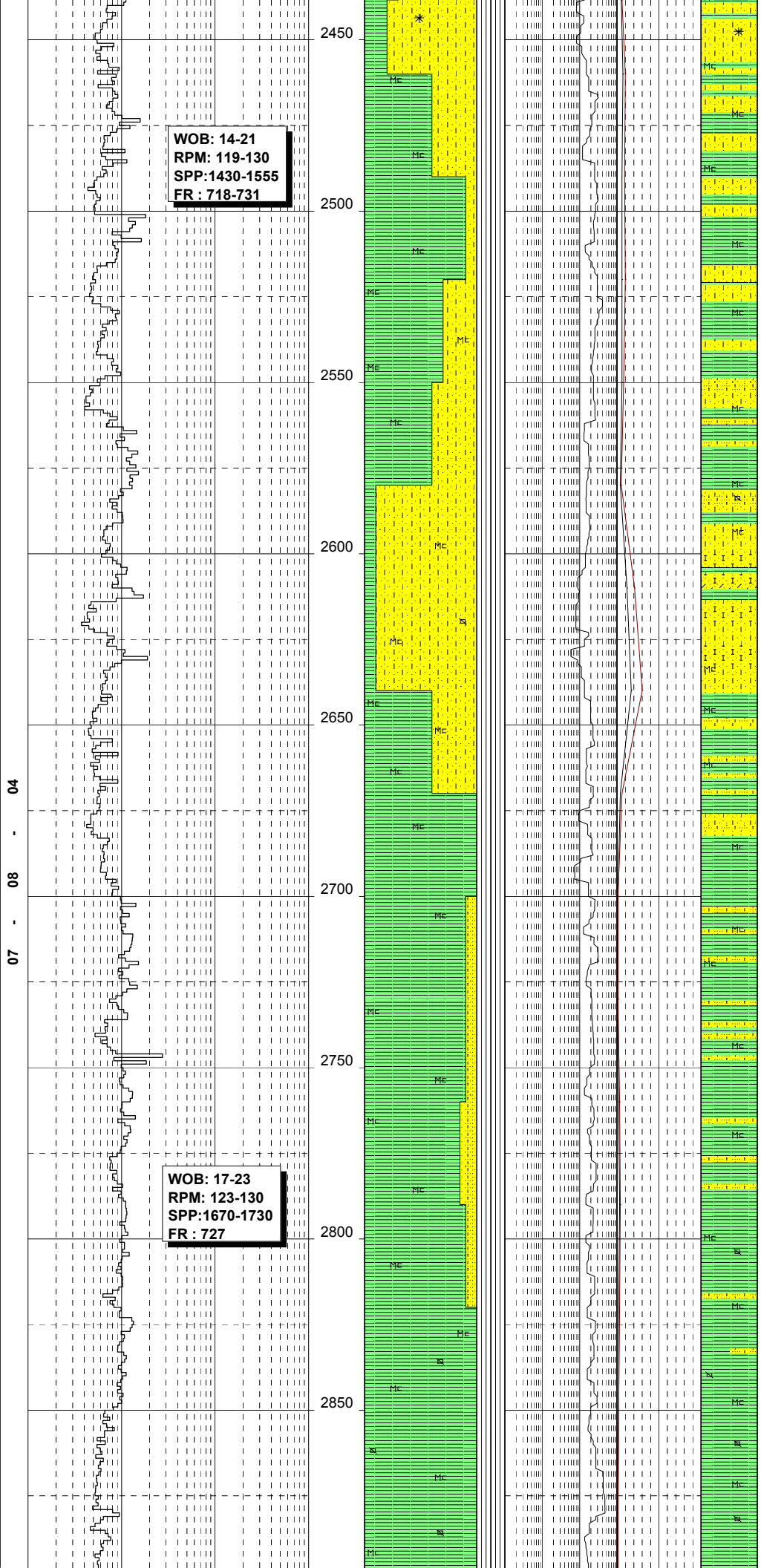
From 2183 ft to 2207 ft
SILTSTONE:grey green,dark green,dark brown,blackish,medium hard,highly argillaceous,pyritic,highly glauconitic, grading to silty Shale.
SANDSTONE:as above.
Thin streaks of
LIMESTONE:a.a. & grey brown i.p,medium hard,fine crystalline,argillaceous, bioclastic.

From 2207 ft to 2235 ft
SANDSTONE:cream,off white,friable-medium hard,fine upper grained,subrounded-subang consol w/calcareous cement,glauconitic, bcm very fine grained,friable,slightly argillaceous.
SILTSTONE:green,grey green,friable, argillaceous,highly glauconitic,sandy.
LIMESTONE:cream,light buff,hard,micro cryptocrystalline,bioclastic i.p(Echinoids, shell fragments)

From 2235 ft to 2310 ft
CLAYSTONE/SHALE:light grey,firm,subblocky highly silty,fine disseminated and free pyrite grading to Siltstone,becoming grey grey brown,finely micaceous.
DOLOMITE:light tan,cream,medium hard, micro-very fine crystalline,slightly calcareous,occ light grey & argillaceous
SILTSTONE:off white,light grey,friable argillaceous.

From 2310 ft to 2430 ft
SANDSTONE: brownish, bituminous (blackish,dark brown pin points of bitumen coating quartz grains) & dirty at top, becoming off white,light cream, transparent,fine upper-medium grained, well sorted,moderately-well consolidated by calcareous cement,occasionally poor silica cement,subrounded-subangular, locally micaceous & argillaceous,no shows.
SHALE:grey,grey brown,subblocky-subfissile i.p,micaceous,silty,non-slightly calcareous grading to Siltstone or Claystone i.p.

From 2430 ft to 2580 ft
SHALE:grey-dark grey,grey brownish,firm,

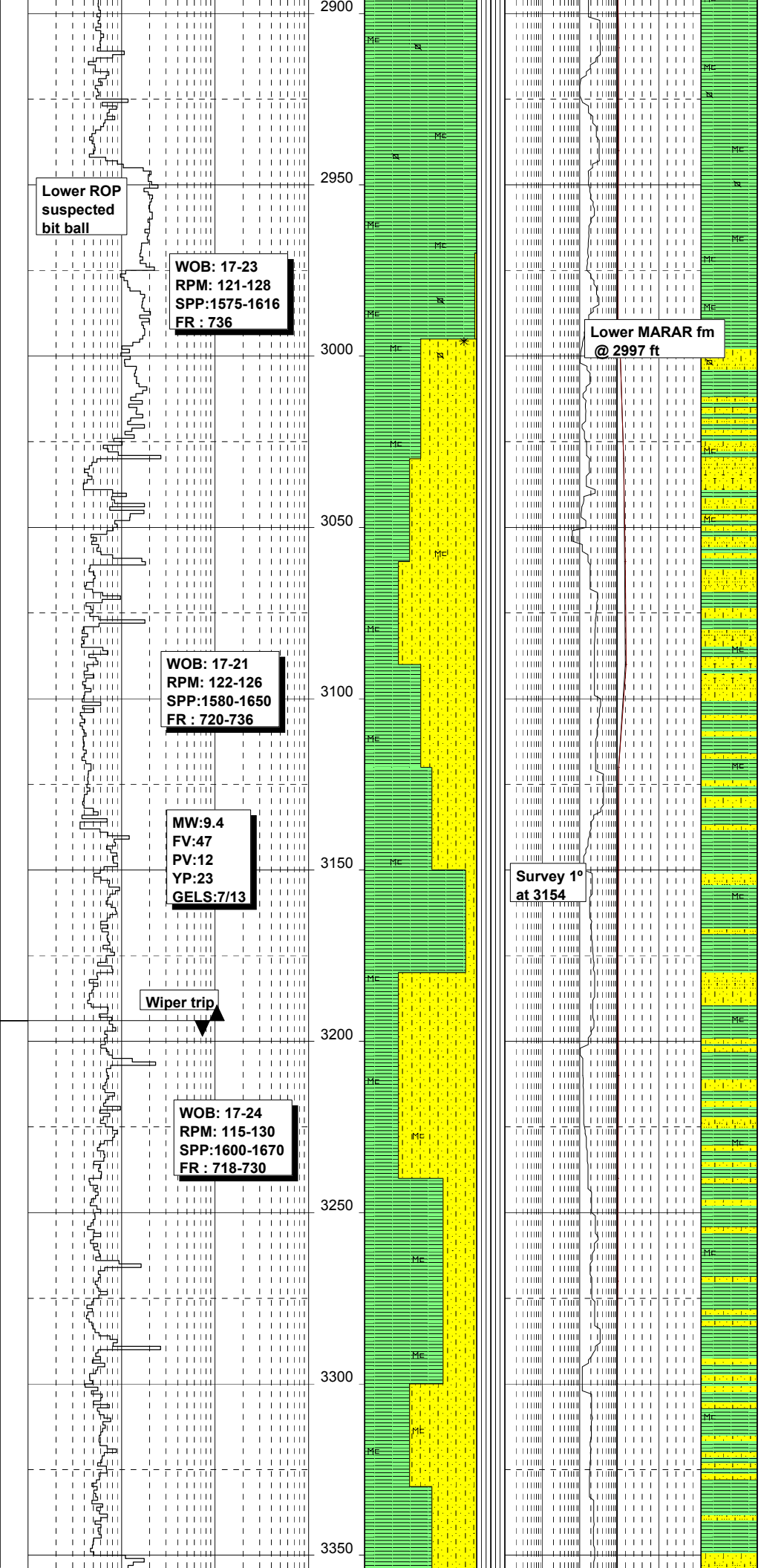


subfissile,occasionally light grey & subblocky,micaceous,slightly calcareous i.p,locally silty.
SANDSTONE:off white,light cream,friable, fine upper grained,well sorted,subrounded subangular,poorly consolidated with calcareous cement,locally very fine grained,argillaceous,highly micaceous, grading to Siltstone,locally glauconitic.

From 2580 ft to 2670 ft
SANDSTONE:a/a,becoming transparent translucent,poorly cemented then well consolidated with dolomitic cement, locally very fine,argillaceous,micaceous grading to Siltstone,pyritic i.p.
SHALE: as above.
Traces of
DOLOMITE: white, medium hard, recrystallised.

From 2670 ft to 2820 ft
SHALE: grey, grey brown, soft-firm, subblocky,highly micaceous,non calcareous washable i.p.w/ minor streaks of
SILTSTONE/SANDSTONE:light grey,creamy, friable-medium hard,slightly-highly argillaceous,micaceous grading to Shale.

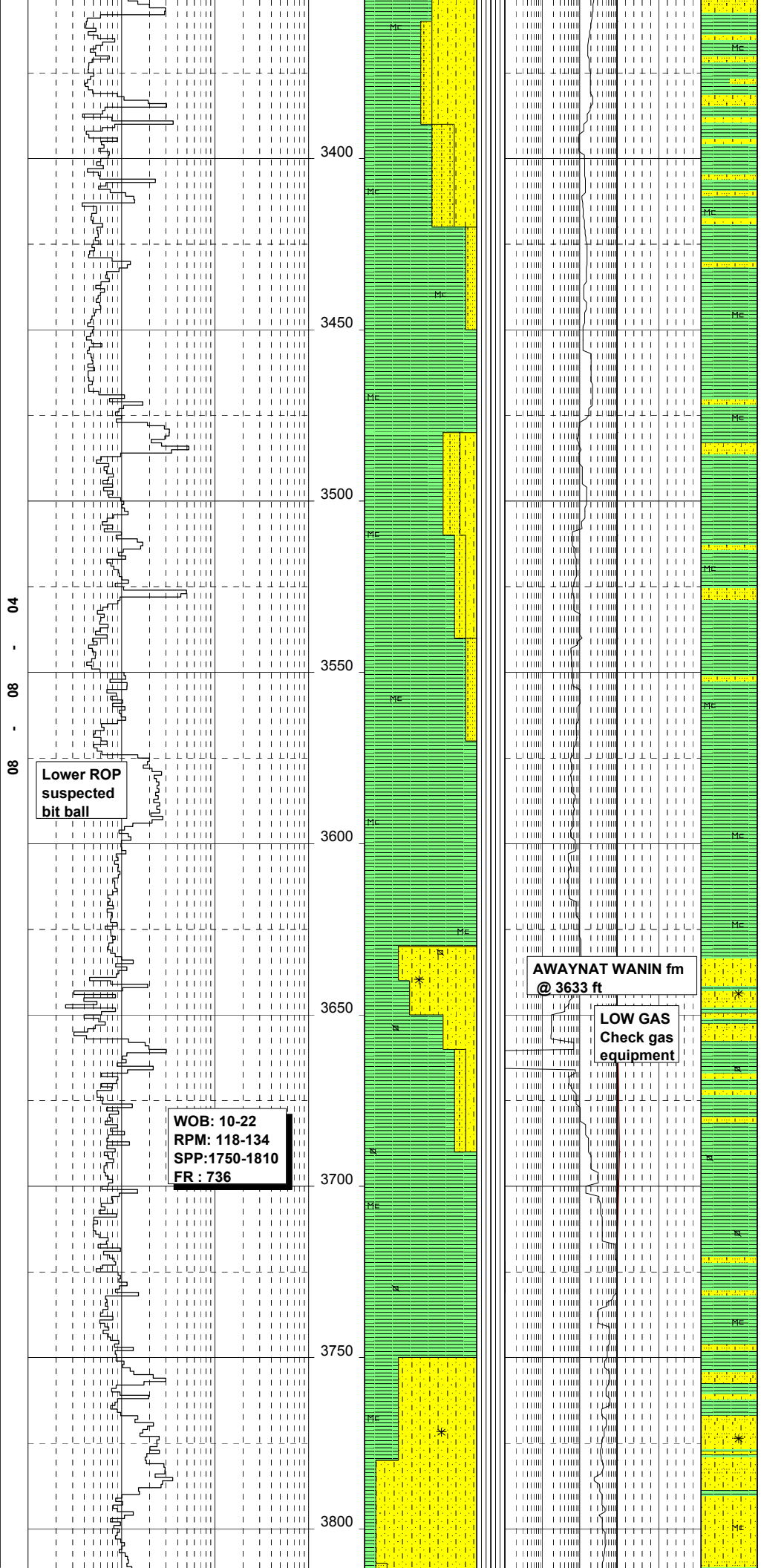
From 2820 ft to 2997 ft
SHALE:grey,dark grey,firm,subfissile, highly finely micaceous,highly pyritic i.p.,common pyritic clusters,non calcareous
with traces of
SILTSTONE/SANDSTONE:as above.



From 2997 ft to 3090 ft
SANDSTONE:off white,very light grey, medium hard,very fine grained,subrounded-subangular,very well consolidated w/ silica cement,slightly calcareous,common pyritic cement at top,slightly argillaceous i.p.,locally glauconitic & micaceous grading to Siltstone.
SHALE:as above locally brownish,highly micaceous.

From 3090 ft to 3330 ft
SHALE:grey,dark grey, firm,subfissile, finely micaceous,no calcareous.
SANDSTONE:very light grey,locally off white,friable,very fine grained, consolidated with silico-argillaceous cement,no calcareous,sligtly micaceous.

From 3330 ft to 3570 ft
SHALE:grey,firm,subfissile,micaceous-highly micaeous,non calcareous,becoming from 3420 ft mainly dark grey,fissile i.p
SANDSTONE:off white,very light grey, friable-medium hard,very fine-fine grained



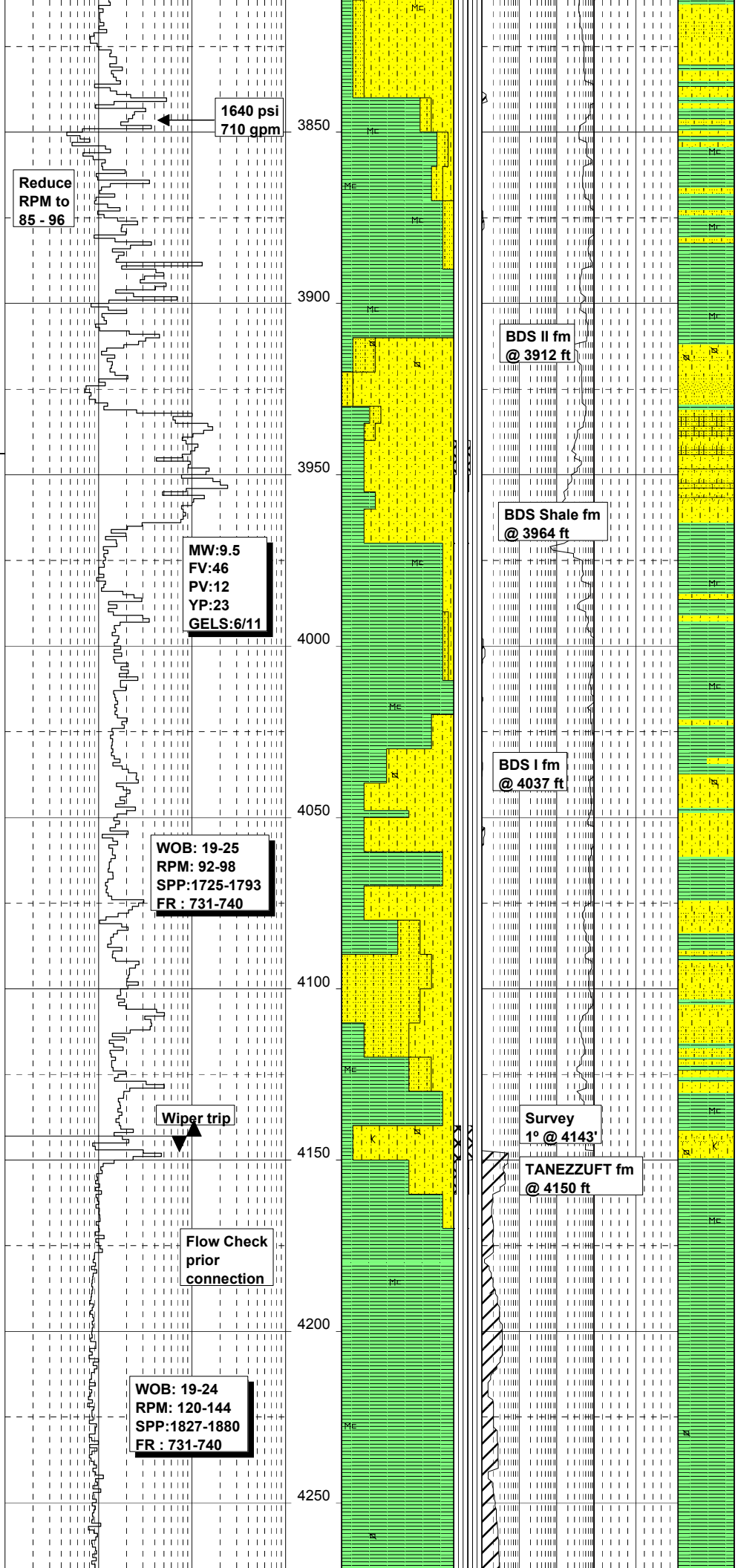
consolidated w/ silico-argillaceous cement, slightly argillaceous,micaceous i.p., very slightly glauconitic.
SILTSTONE:light grey,medium hard argillaceous.

From 3570 ft to 3633 ft
SHALE: as above.

From 3633 ft to 3660 ft
SANDSTONE:transparent,translucent,tanish, off white,light grey,common loose,friable fine upper-medium grained,poorly consol w/ silica cement,locally medium hard & slightly dolomitic,subrounded-subangular, occ angular,common pyritic cluster & cement,occurence of oolithic streaks w/ argillaceous matrix,fair visible intergranular porosity,no shows.
SHALE:as above & grey brownish,micaceous & pyritic,silty i.p. grading to SILTSTONE.

From 3660 ft to 3750 ft
SHALE:grey,dark grey,firm,subfissile-fissile,micaceous,finely pyritic,silty i.p.

From 3750 ft to 3840 ft
SANDSTONE:off white,transparent,friable, medium hard,fine upper-medium grained, subrounded-subangular,well consolidated w/silica cement,occ micaceous,few scattered glauconitic grains,no visible porosity,no shows.
SHALE:as above, becoming splintery i.p.



From 3840 ft to 3912 ft
SHALE:mainly brown,brownish grey,firm, subfissile,highly micaceous,no calcareous silty i.p. w/ laminations of SANDSTONE:a/a argillaceous,micaceous & SILTSTONE:light grey,light green, argillaceous,friable.

From 3912 ft to 3964 ft
SANDSTONE:mainly off wh,lt gry at top,fri occ hd w/ Pyr cmt,f up gr,occ med gr, subrnd,consol w/abd sil-arg cmt,bcm loc transp-transl,lse,med-crs gr & rnd,then f/3933 ft,qtz,v hd w/ transp intergran sil cmt,no vis Por,only few cuttings (3940'-3950') v slily brnsh give yel-gold yel dir Fluor,med fast milky yel cut & fnt r.r.
Minor SHALE:gry-dk gry,firm,subblky-subfis, no calc,finely Pyr and SILTSTONE:brwn,med hd-hd,arg,grad to Shale.

From 3964 ft to 4037 ft
SHALE:grey,occ grey brown,firm,subfissile fissile,no calcareous,finely micaceous, with some laminations of SANDSTONE/SILTSTONE:off white,light grey friable-medium hard,very fine grained, consolidated w/ silica cement, argillaceous.

From 4037 ft to 4060 ft
SANDSTONE:translucent,very light cream, friable,occ medium hard,very fine grained subrounded-subangular,consolidated w/ silica cement,bcm w/ depth off white,light grey & argillaceous,pyritic i.p.,no visible porosity,no shows.
SHALE:light grey,grey,firm-medium hard, locally micaceous,silty.

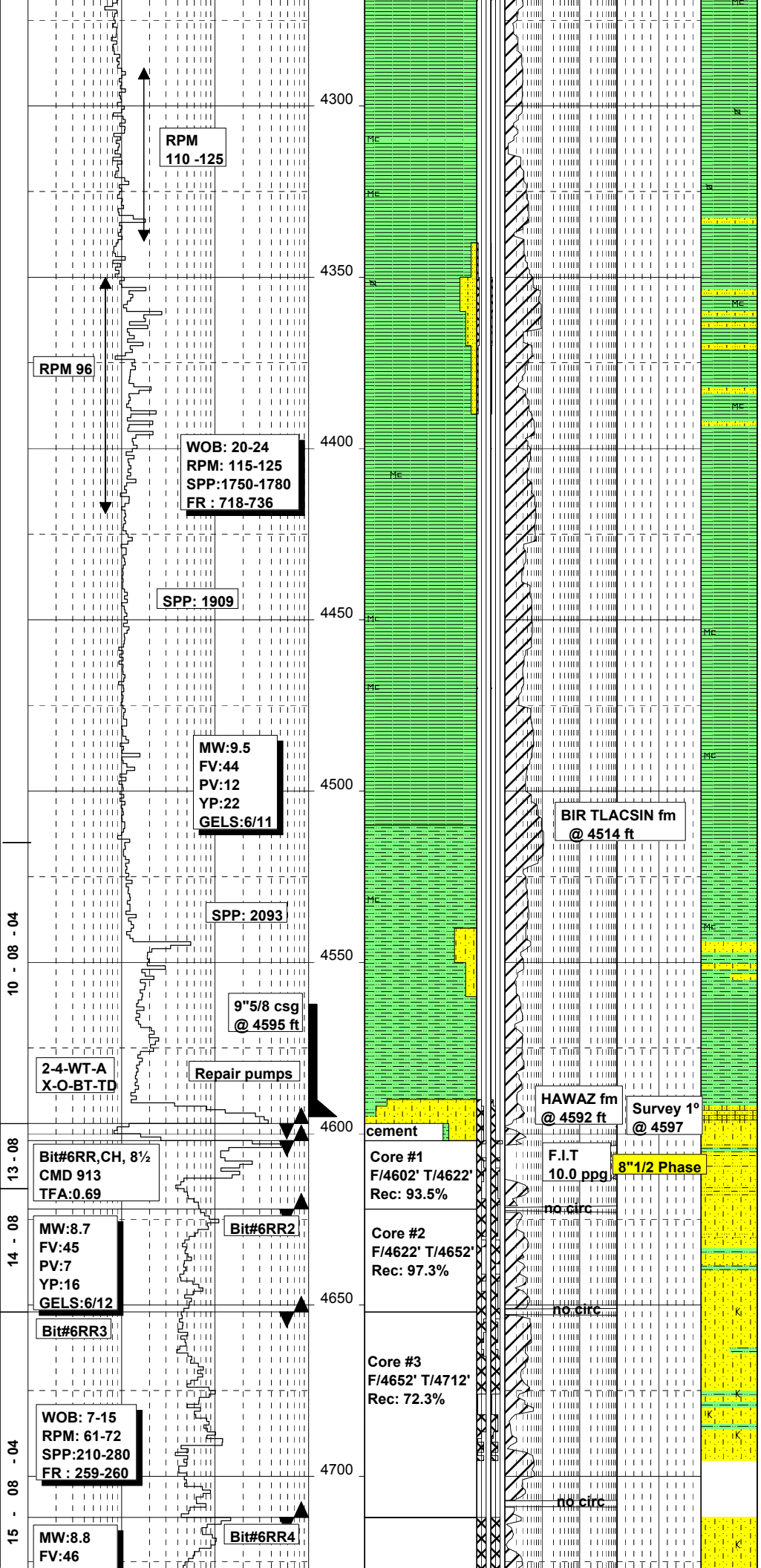
From 4060 ft to 4080 ft
SANDSTONE:a/a,mainly off white,bcm medium hard.
SHALE:grey,slightly brownish,firm,fissile subfissile,splintery i.p,slightly micaceous grading to Siltstone.

From 4080 ft to 4130 ft
SILTSTONE:grey brown,medium hard,argilla-ceous,micaceous,sandy grad to Sandstone.
SANDSTONE:a/a.
SHALE:grey,dark grey,firm,subfissile-fissile,micaceous,finely pyritic silty i.p.

From 4130 ft to 4143 ft
SANDSTONE:off wh,transp,fri,med hd,f up-med gr,subrnd-subang,w consol w/sil cmt, occ mic,few scat glauc gr,no vis Por,N/S SHALE:a/a bcm splint i.p.

From 4143 to 4150 ft
SANDSTONE:transp,transl,v lt brn,med hd-fri,med gr,occ f or crse gr,subrnd-subang poorly cmt w/sil cmt,loc kaol or qtz,bcm lse & badly sort,poor-loc good intgran Por v lt brn oil stn few amber oil droplets, gd brt yel dir Fluor,slw milky wh cut Fluor,wk brt yel r.r.

From 4150 ft to 4350 ft
SHALE:brownish grey,firm,subfissile-fissile,micaceous,finely pyritic.



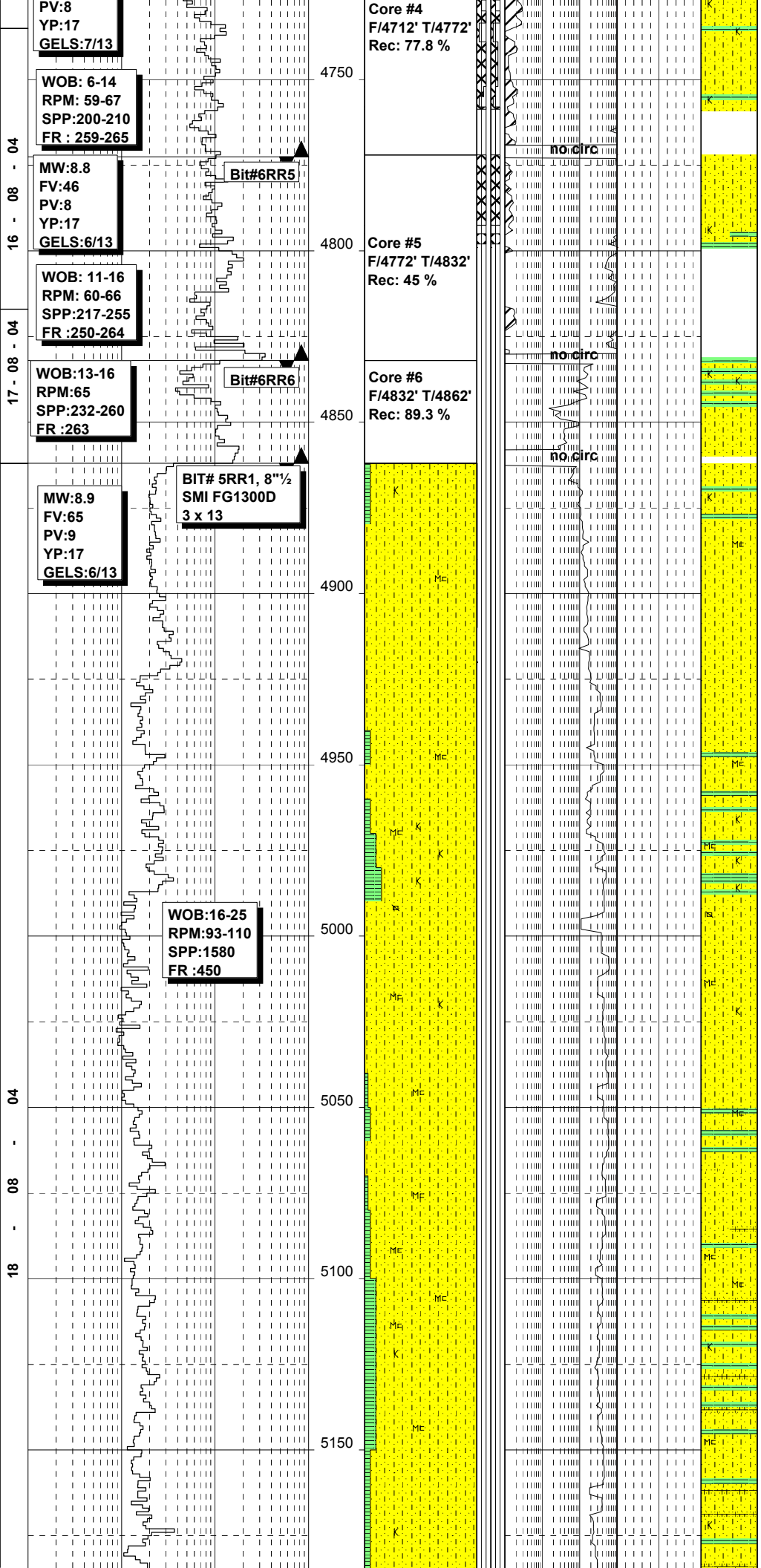
From 4350 ft to 4390 ft
SILTSTONE:brownish,medium hard-hard, micaceous i.p.,finely pyritic,grading to very fine Sandstone w/ bright yellow direct Fluor,slow bluish white cut Fluor.
SHALE:as above.

From 4390 ft to 4514 ft
SHALE:as above,becoming from 4450 ft grey,firm,fissile,splintery,micaceous, occasionally silty

From 4514 ft to 4592 ft
SHALE:grey brown,brown,firm-medium hard, subblocky,w/ scattered medium-coarse rounded quartz grains,silty becoming highly silty w/ depth grading to Siltstone.
with streaks of :
SANDSTONE:off white,light grey,hard-very hard,very fine grained,consolidated with silica cement,slightly micaceous,no porosity,no shows.

From 4592 ft to 4602 ft
SANDSTONE:transp,transl,appears mainly as lse Sand,v hd(rock flour),med-crs gr,rnd, occ broken gr,transp sil cmt,qtz,some gr coated w/ bit pin points,occ pyr,n vis Por,v lt brn O stn,occ O droplets,fr brt yel dir Fluor,wk slow wh yel cut,wk brt yel r.r.

From 4602 ft to 4862 ft.
Refer to core descriptions.

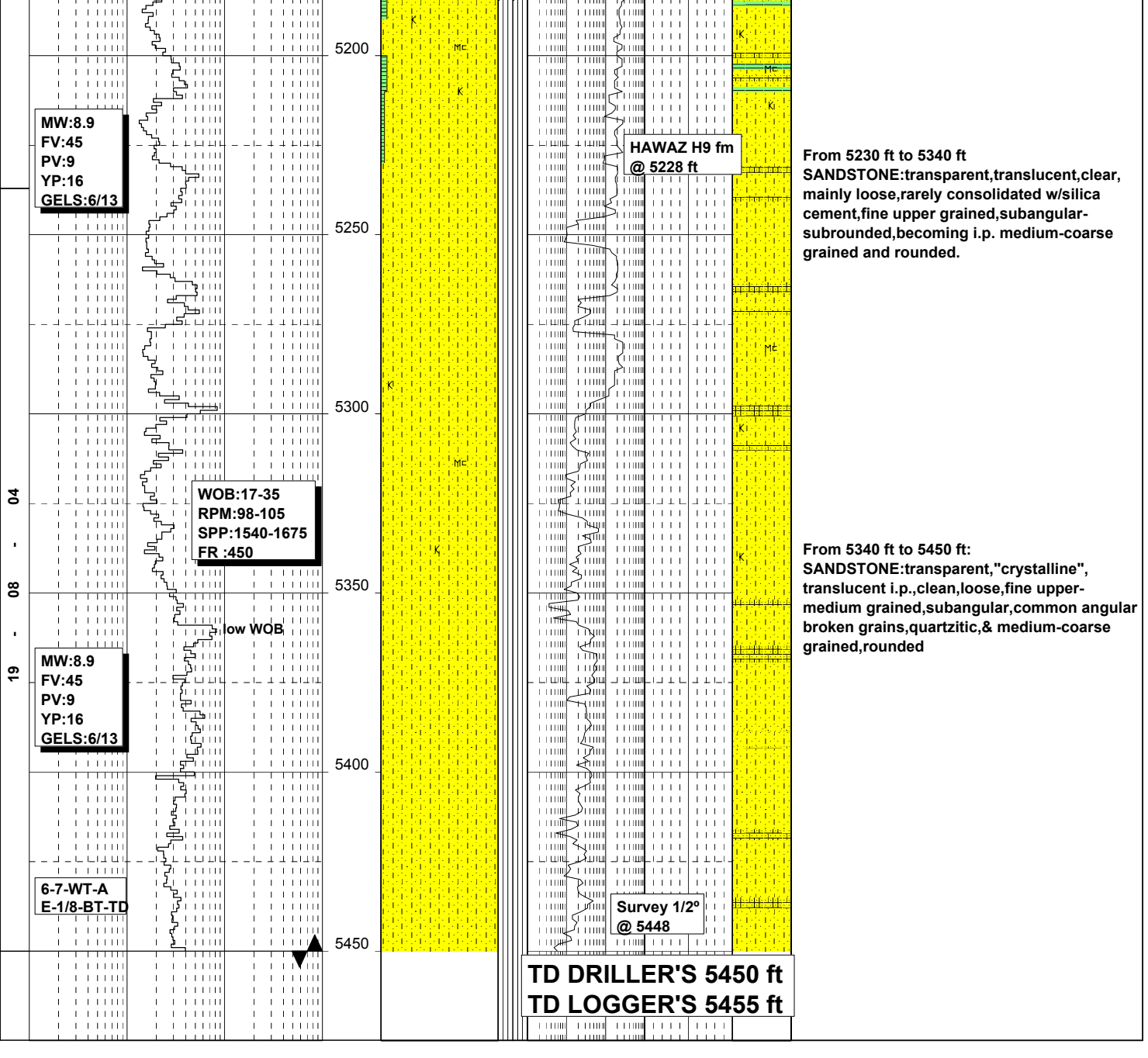


From 4862 ft to 4930 ft
SANDSTONE:transparent,clear,appears mainly loose @ surface,poorly consolidated w/ silica cement,fine upper grained, subrounded,few scattered mica,quartzitic i.p. w/intergrown transparent silica cmt, occ kaolinite,abundant rock flour i.p, few % cuttings give brigth yellow direct fluorescence,with traces of SHALE:brown,grey brown,firm,subfissile-fissile,micaceous,silty i.p,or black, redish,fissile,with iron oxyds.

From 4930 ft to 5000 ft
SANDSTONE:as above,becoming translucent, off white,medium hard,fine upper grained, consolidated with silica cement,slightly micaceous,slightly-fairly kaolinitic, poor visible porosity,occasional pyrite clusters,with traces of black brownish dead oil stains.
Minor SHALE as above.

From 5000 ft to 5080 ft
SANDSTONE:a/a,becoming very light brown no to slightly kaolinitic,slightly micaceous.
with traces of SHALE:brown,subfissile,highly micaceous, silty-sandy i.p.

From 5080 ft to 5233 ft:
SANDSTONE:off white,transparent-translucent,mainly loose,but friable-medium hard,fine grained,subrounded-subangular, consolidated with poor silica cement, common kaolinitic matrix,locally slightly quartzitic,no-very poor visible porosity, no shows, with
SHALE: as above.



From 5230 ft to 5340 ft
SANDSTONE:transparent,translucent,clear, mainly loose,rarely consolidated w/silica cement,fine upper grained,subangular-subrounded,becoming i.p. medium-coarse grained and rounded.

From 5340 ft to 5450 ft:
SANDSTONE:transparent,"crystalline", translucent i.p.,clean,loose,fine upper-medium grained,subangular,common angular broken grains,quartzitic,& medium-coarse grained,rounded